Solving quadratic equations

1

a $y = 0, \frac{1}{5}$ **b** $x = -11, 5$ **c** $x = \frac{1 \pm \sqrt{161}}{8}$ **d** $m = 4 \pm \sqrt{10}$

e $x = 9, -5$ **f** $x = -\frac{2}{3}, 5$

2

a Yes **b** No **c** Yes **d** No

3

a $x = -4, -5$ **b** $x = -4, -5$

Applications

4 $x = 3, -7$

5

a 313.6 metres **b** $t = 4$

6

a $t = 3$ **b** $t = 9$

7 $t = 6$

8

a $m = 1900$ **b** $64.8\ \text{km/h}$ **c** $v = 16$

9

a $S = 528$ **b** $n = 5$

10 $n = 40$

11

a $x = 7$ **b** 8

12 $x = 7, -8$

13

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a $t = 2, 3$ **b** 5 sec **c** No

14
a $t = \frac{1}{4}, 2\frac{3}{4}$ **b** $2\frac{1}{2}$ sec **c** $t = 1\frac{1}{4}, 1\frac{3}{4}$ **d** $\frac{1}{2}$ sec

15
a $y = 14 - x$ **b** $x = 8, 6$ **c** $y = 6, 8$

16
a $w = 10$ **b** 16 metres

17 $t = 4$

18
a $x = 10$ **b** 23 metres

19
a $x = 9$ **b** 19 **c** 21

20 $x = 10$

21
a $r = 6$ **b** 6.6 m

22
a $4x^2 + 44x$ m^2 **b** $x = 2$

23
a $m = 10$ **b** 15 hours

24
a $w = 2.33$ **b** 15.3 cm

25
a 41 **b** $70 - x$ **c** $70x - x^2$ **d** 1225
e Yes

26
a $x = 5, -1$ **b** $(5, 7), (-1, 7)$

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27 $x = 8$

28

a $x^2 + 7x \text{ m}^2$

b $x^2 + 23x + 120 \text{ m}^2$

c $x = 23$

d Width $= 23$ m, Length $= 30$ m

Further quadratic equations

29 $x = 0, 3$

30

a $\dfrac{x^2 + y^2}{xy}$

b $xy = 12$

c $x = -4, -3, 3, 4$

d $y = -3, -4, 4, 3$

e $\dfrac{3}{4}$

31

a **C**

b $x = \dfrac{-1 + \sqrt{5}}{2}$

32 $x = \dfrac{1 + \sqrt{33}}{2}$

33

a $x = -1, -4$

b $x = 2, -3, -5$

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